

A Stability-Aware Framework for Early Prediction of Signal Degradation and Handover Triggering in 5G Networks

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Abstract— Efficient mobility management remains a key challenge in 5G and beyond networks, especially in high-speed and dense deployment scenarios. Traditional handover mechanisms based on static RSRP thresholds are reactive and often delay handover decisions, increasing user exposure to poor signal conditions. This paper proposes a machine learning-based proactive handover framework that predicts signal degradation 3 seconds in advance using a stability-aware composite risk score (R_t). A HistGradientBoostingClassifier is trained on 103,542 real-world samples from the UCC Cork dataset, with 62 engineered temporal and statistical features. The proposed risk formulation integrates smoothing and volatility damping to suppress unstable triggers. The model achieves a ROC-AUC of 0.985 on a chronological test set. Compared to raw model output (P_t), the proposed R_t reduces false handover alarms by 52.6% and improves F1-score, with a minor reduction in recall. Simulation results show approximately 60% reduction in time spent under degraded signal conditions compared to traditional methods, without increasing unnecessary handovers. The results demonstrate the effectiveness of stability-aware, ML-driven handover strategies for next-generation wireless networks.

Index Terms— 5G Handover, Signal Degradation Prediction, HistGradientBoosting, Exponential Moving Average, Volatility Damping, Proactive Mobility Management, RSRP, UCC Cork Dataset, Risk Score, Machine Learning.

I. INTRODUCTION

The rapid evolution of fifth-generation (5G) cellular networks has fundamentally altered the landscape of wireless mobility management. With peak data rates exceeding 10 Gbps, sub-millisecond latency targets, and inter-site distances as small as 100 meters in dense urban deployments [7], the frequency and criticality of handover

decisions have increased dramatically compared to previous generations. In vehicular and high-mobility contexts, a user equipment (UE) may traverse multiple cells within seconds, placing enormous pressure on the handover mechanism to act swiftly and accurately.

The 3GPP standard handover procedure (TS 38.300, TS 38.331) [9][10] relies primarily on A3 event triggering: a handover is initiated when the target cell's RSRP exceeds the serving cell's RSRP by a configurable hysteresis margin for a time-to-trigger (TTT) duration. While this mechanism is robust and well-studied, it is fundamentally reactive—the UE must already be experiencing relative signal decline before any action is taken. In fast-fading channels or high-speed scenarios, the TTT window may expire only after the signal has already degraded below acceptable thresholds, causing packet loss, increased retransmissions, and degraded quality of experience.

Machine learning offers a compelling alternative paradigm: rather than reacting to observed degradation, a trained model can anticipate future signal quality from temporal patterns in current and historical KPIs. However, deploying raw ML classifier outputs directly as handover triggers introduces new problems: high output variance in noisy radio environments leads to oscillatory trigger behavior, known as ping-pong handover, which generates excessive signaling overhead and degrades network throughput [4].

This paper addresses both challenges through a three-stage temporal risk pipeline. First, a HistGradientBoostingClassifier generates a per-timestep degradation probability P_t from 62 engineered features derived from real-world 5G drive-test data. Second, an exponential moving average (EMA) produces a smoothed trend signal S_t that suppresses short-duration P_t spikes. Third, a volatility term V_t —computed as the mean absolute

deviation of recent P_t values—modulates the composite risk score R_t , reducing sensitivity during periods of unstable prediction. A handover is triggered only when R_t exceeds a configurable threshold and the target cell is measurably stronger than the serving cell.

The key contributions of this work are:

1. A stability-aware risk scoring pipeline ($P_t \rightarrow S_t \rightarrow V_t \rightarrow R_t$) that improves handover decision quality over raw ML output.
2. Training and rigorous evaluation of a HistGradientBoostingClassifier on 103,542 real-world 5G measurements from the UCC Cork dataset, using a leakage-free chronological train/test split.
3. A multi-threshold sweep across nine operating points (0.30–0.70) demonstrating that R_t consistently dominates P_t in precision and F1-score at every threshold.
4. A real-time inference benchmark confirming 81× headroom against a 1-second tick budget (12,402 μ s per inference vs. 1,000,000 μ s budget).
5. Simulation results demonstrating approximately 60% reduction in time spent below the -100 dBm degradation threshold compared to traditional RSRP-based handover.

II. RELATED WORK

Handover optimization has attracted significant research attention across all generations of cellular networks. Early work focused on adaptive hysteresis and time-to-trigger parameter tuning [4] to reduce ping-pong events. More recently, machine learning has emerged as a dominant paradigm for proactive mobility management.

Hatipoğlu et al. [1] proposed gradient boosting and neural network models for handover prediction in beyond-5G networks, demonstrating improved prediction accuracy over conventional methods. Lima et al. [2] investigated recurrent architectures including LSTM networks for sequence-aware handover decisions, highlighting the importance of temporal feature extraction. Dzaferagic et al. [3] implemented ML-based handover prediction directly within a real Open RAN (O-RAN) deployment using the RAN Intelligent Controller (RIC) xApp interface, confirming practical feasibility. Karmakar et al. [4] proposed adaptive time-to-trigger and hysteresis margin mechanisms controlled by an RL agent, reducing unnecessary handovers in dense heterogeneous networks. Rajesh et al. [5] demonstrated AI-based handover triggering with significant gains in ultra-dense network scenarios, and Zaid et al. [6] applied ML to cognitive Wi-Fi AP selection in high-mobility environments.

Despite this progress, several important gaps persist:

Gap 1: Most works optimize for prediction accuracy (ROC-AUC or recall) without explicitly modeling decision stability. Raw ML outputs applied directly as triggers suffer from chattering in noisy radio conditions. Gap 2: Volatility as an explicit input to the handover decision function has not been previously formalized. Gap 3: Systematic multi-threshold evaluation comparing raw model output against a smoothed risk score is absent from prior literature. Gap 4: Real-world dataset validation with rigorous chronological splitting to prevent temporal leakage is inconsistently applied (Table I).

TABLE I: Comparison with Prior Methods

Method	Accuracy Focus	Stability Focus	Real Dataset	Multi-Threshold	Leakage-Free
Hatipoğlu et al. [1]	Yes	No	Yes	No	Partial
Lima et al. [2]	Yes	No	Yes	No	No
Dzaferagic et al. [3]	Yes	No	Yes (O-RAN)	No	Yes
Karmakar et al. [4]	Yes	Partial (RL)	Yes	No	Partial
This Work	Yes	Yes (Vt)	Yes (UCC)	Yes (9 pts)	Yes

III. PROBLEM STATEMENT

Let a UE be connected to a serving gNB and moving through a coverage area. At each discrete time step t , the network measures a vector of radio KPIs $x_t = [\text{RSRP}_t, \text{RSRQ}_t, \text{SNR}_t, \text{CQI}_t, \text{RSSI}_t, v_t, \text{rd}_t, r_t^U]^T$ plus derived features. The goal is to determine, at each t , whether the UE should be handed over to a target gNB before the serving signal degrades below a critical threshold $\gamma = -100$ dBm.

The binary label is defined as: $Y_t = 1$ if $\text{RSRP}_{t+k} < \gamma$, where $k = 3$ seconds is the prediction horizon. A handover triggered at time t is considered proactive if it occurs before the degradation event, and reactive if it is triggered after or coincident with crossing γ .

The dual-objective optimization problem is therefore:

$$\text{maximize } \text{Recall}(Y, HO) \times \text{Precision}(Y, HO)$$

subject to: (i) the trigger signal HO_t is temporally stable (low variance), and (ii) the inference latency satisfies the 1-second real-time tick budget. This formulation captures the fundamental tension between sensitivity (catching all

degradation events early) and specificity (not triggering unnecessary handovers that cause signaling overhead and brief service interruption).

IV. DATASET AND PREPROCESSING

A. Dataset Description

The UCC Cork mobility dataset comprises real-world 5G network measurements collected during vehicular drive-tests across the Cork metropolitan area, Ireland. The dataset contains measurements from 60 distinct mobility traces (trace IDs) spanning diverse road types, speeds, and signal conditions. After loading and removing corrupted entries (RSRP = -200 dBm sentinel values), the working dataset contains 103,542 rows with 27 raw columns.

Key raw features include:

- RSRP (Reference Signal Received Power, dBm) — primary signal strength indicator
- RSRQ (Reference Signal Received Quality, dB)
- SNR (Signal-to-Noise Ratio, dB)
- CQI (Channel Quality Indicator, 1–15)
- RSSI (Received Signal Strength Indicator, dBm)
- NRxRSRP (neighboring cell reference signal power)
- UE speed (m/s)
- DL/UL bitrate (Mbps)

B. Label Generation

A binary degradation label Y_t is created per trace using a $k = 3$ -second look-ahead: $Y_t = 1$ if the RSRP measured 3 rows later (within the same trace) falls below -97 dBm. The shift is applied exclusively for label generation and never used as an input feature, ensuring strict temporal leakage prevention. The resulting class distribution is 81.9% negative (no degradation) and 18.1% positive (impending degradation), motivating the use of class-balanced training.

C. Feature Engineering

Per-trace feature engineering produces 62 derived features from the 5 primary signal columns (RSRP, RSRQ, SNR, CQI, RSSI) plus throughput and speed. The engineering pipeline includes: (i) lag features at offsets 1, 2, 3 timesteps (backward-only, no future leakage); (ii) first and second discrete derivatives (Δx_t , $\Delta^2 x_t$); (iii) rolling mean, minimum, and standard deviation over a 5-sample window with $\text{min_periods}=1$; and (iv) RSRP proximity margin to the -97 dBm threshold. All rolling operations use left-anchored windows. The 100.0% row retention rate ($103,545 \rightarrow 103,542$, loss of 3 header artefacts only) confirms no spurious row loss from feature engineering.

D. Train/Test Split

To prevent temporal leakage between traces, a chronological group split is applied: the 60 traces are ordered by their earliest timestamp, and the first 70% (42 traces, 72,492 rows) are assigned to training while the remaining 30% (18 traces, 31,050 rows) form the test set. This produces a naturally harder test set, as seen in the class-1 rate dropping from 21.1% in training to 11.1% in test—reflecting temporal non-stationarity in real drive-test data.

TABLE II: Dataset Split Summary

Split	Traces	Rows	Class-1 Rate	Role
Training	42	72,492	21.1%	Model fitting
Test (held-out)	18	31,050	11.1%	Evaluation / risk demo
Total	60	103,542	18.1%	UCC Cork dataset

V. PROPOSED METHODOLOGY

The proposed framework operates as a layered temporal inference pipeline, illustrated in Fig. 1. At each 1-second tick, raw KPI measurements from the serving gNB are fed into the feature engineering module, which reconstructs the 62-dimensional feature vector using only past observations. The engineered vector is passed to the trained classifier, which outputs P_t . The temporal risk module then computes S_t , V_t , and R_t in sequence before issuing a binary handover recommendation.

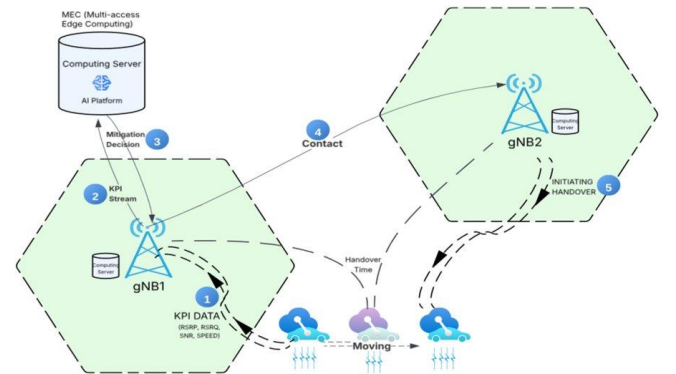


Fig. 1. System architecture of the proposed proactive handover framework. The UE streams KPI data (RSRP, RSRQ, SNR, Speed) to gNB1, which forwards it to the MEC-hosted AI platform. The ML model evaluates degradation risk and issues a mitigation decision. Contact with gNB2 is established and handover is initiated before signal collapse.

A. Degradation Prediction Model (P_t)

A `HistGradientBoostingClassifier` (scikit-learn [12]; conceptually equivalent to LightGBM [11]) is selected as the primary model for three reasons: native handling of missing values via histogram-based binning, native class-weight balancing, and strong empirical performance on tabular KPI data. The model is configured with `max_iter = 500`, `learning_rate = 0.05`, `max_depth = 6`, `min_samples_leaf = 25`, `l2_regularization = 0.1`, and `class_weight = 'balanced'`. The output $P_t \in [0, 1]$ represents the estimated probability that RSRP will fall below -97 dBm within the next 3 seconds.

B. Temporal Smoothing via EMA (St)

To suppress high-frequency oscillations in P_t , an Exponential Moving Average is applied at each tick:

$$S_t = \alpha \cdot P_t + (1 - \alpha) \cdot S_{t-1}$$

where $\alpha = 0.5$ is the smoothing factor. The EMA is stateful across ticks within a session and reset to zero at session initialization. S_t captures the medium-term degradation trend, filtering out momentary spikes caused by fast-fading or measurement noise.

C. Volatility Damping (Vt)

The volatility term $V_t \in [0, 1]$ quantifies the instability of recent classifier predictions over a sliding window of $N = 5$ timesteps:

$$V_t = \min \left[\frac{\langle |\Delta P| \rangle_{2-5}}{1.0}, 1.0 \right]$$

where $\langle |\Delta P| \rangle$ is the mean of absolute consecutive differences of P_t over the window. When P_t fluctuates rapidly—indicating an unreliable signal environment— V_t approaches 1.0 and suppresses R_t toward zero, preventing hasty handover decisions. Conversely, when P_t evolves smoothly, V_t remains near zero and allows R_t to respond to genuine degradation trends.

D. Composite Risk Score and Handover Decision (Rt)

The composite risk score is defined as:

$$R_t = (0.4 \cdot P_t + 0.4 \cdot S_t) \cdot (1 - V_t)$$

The equal weighting of P_t and S_t ensures that both the instantaneous prediction and the trend contribute symmetrically, while the $(1 - V_t)$ multiplicative term scales the score down during volatile periods. A proactive handover is triggered at time t if and only if:

$$R_t > \tau \text{ AND } RSRP_{target} > RSRP_{server}$$

where the default threshold $\tau = 0.55$. The second condition ensures handover is initiated only when a measurably stronger target cell is available, preventing unnecessary handovers in multi-path fading scenarios.

VI. EXPERIMENTAL SETUP

A. Implementation Details

All components are implemented in Python 3.x. The classifier is provided by scikit-learn 1.x [12]. Feature engineering uses NumPy and Pandas. The dataset is loaded from the UCC Cork Excel traces and processed entirely in memory. A custom Streamlit-based simulation dashboard was developed to visualize handover decisions in real time. The real-time inference latency benchmark (5,000 calls on test data) yields 12,402 μ s per timestep—providing an $81\times$ headroom factor against the 1-second (1,000,000 μ s) tick budget, confirming viability for real-time deployment.

B. Experimental Protocol

Evaluation proceeds through five phases:

6. Model Training: `HistGradientBoostingClassifier` fitted on 42 training traces with class balancing.
7. Model Evaluation: ROC-AUC, recall, precision, F1, and confusion matrix on 18 held-out test traces.
8. Feature Importance: Permutation importance (`n_repeats = 10`) on the test set to identify the most predictive KPIs.
9. Risk System Comparison: P_t vs. R_t evaluated side-by-side at nine threshold values from 0.30 to 0.70.
10. Simulation: Custom mobility simulator comparing ML-based and traditional handover over 200–250-second traces.

VII. RESULTS AND DISCUSSION

A. Model Performance

The `HistGradientBoostingClassifier` achieves a ROC-AUC of 0.9851 on the chronological test set, indicating strong discriminative ability across all operating thresholds. At the default threshold of 0.50, the model yields recall of 0.9309 on the degradation class (class 1), with precision of 0.6983 and F1-score of 0.7980. The confusion matrix shows $TN = 26,201$, $FP = 1,391$, $FN = 239$, $TP = 3,219$. The high recall is intentional: the training objective prioritizes catching degradation events early, accepting some false alarms that will be filtered by the volatility damping stage.

A leakage audit confirms the high AUC is genuine: the forward-looking shift used for label creation (shift(-3)) is strictly separated from all input features; lag features use backward shifts only; rolling windows are left-anchored with `min_periods = 1`; and train/test split is chronological with no cross-trace bleed.

B. Feature Importance

Permutation importance analysis (Table III) identifies RSRP as the dominant predictor by a wide margin (importance = 0.0325, approximately 5.6× the second-ranked feature). The neighboring cell power NRxRSRP ranks second (0.00575), followed by the rolling minimum of RSRP (RSRP_rmin, 0.00523). RSSI and its derived features (rolling min, lag features, delta) collectively form the second most important signal group. UE speed (rank 12) and throughput metrics (DL/UL bitrate, ranks 14–15) provide complementary mobility-domain information. Notably, second-order derivative features (delta2_RSSI, rank 23) contribute positively, confirming that acceleration in signal change carries predictive value beyond first-order rate-of-change alone.

TABLE III: Top-15 Features by Permutation Importance (ROC-AUC Drop)

Rank	Feature	Importance	Std Dev
1	RSRP	0.03253	0.00057
2	NRxRSRP	0.00575	0.00027
3	RSRP_rmin	0.00523	0.00016
4	RSSI	0.00270	0.00020
5	RSRP_rmean	0.00132	0.00010
6	RSSI_rmin	0.00068	0.00018
7	RSSI_lag3	0.00062	0.00009
8	RSRP_lag1	0.00035	0.00008
9	delta_RSSI	0.00034	0.00004
10	RSSI_rstd	0.00030	0.00007
11	RSRP_lag3	0.00026	0.00008
12	Speed	0.00024	0.00006
13	DL_bitrate	0.00022	0.00006
14	UL_bitrate	0.00020	0.00005
15	SNR_rmean	0.00014	0.00004

C. RSRP Signal Behavior Over Time

Fig. 2 presents the RSRP signal time series from the simulation, comparing the serving cell signal (red), the smoothed serving RSRP (blue), and the raw measurement trace (orange dotted). The red dashed horizontal line marks the -100 dBm degradation threshold. Four distinct cell-transition events are visible as sharp upward peaks in RSRP corresponding to handovers to stronger cells. In each inter-handover period, the signal gradually degrades toward and in some cases crosses the threshold—exactly the conditions the ML pipeline is trained to anticipate.

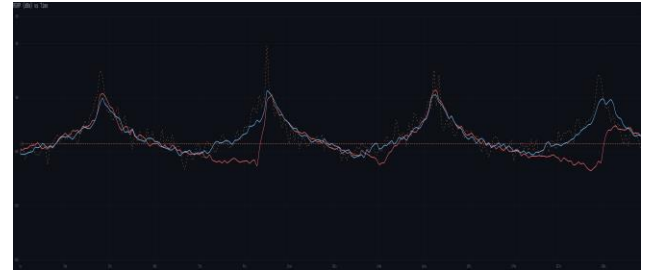


Fig. 2. RSRP (dBm) vs. time during simulation. Red solid: serving cell RSRP. Blue solid: smoothed RSRP. Orange dotted: raw measurement. Dashed red line: -100 dBm degradation threshold. Sharp upward transitions correspond to proactive ML-triggered handover events.

D. Risk Score Temporal Behavior

Fig. 3 shows the temporal evolution of all three pipeline components— P_t (blue), S_t (orange), and R_t (purple)—over the simulation duration. The dashed horizontal line marks the trigger threshold $\tau = 0.55$. Several key behaviors are observable. First, P_t exhibits the highest variance, frequently spiking above the threshold during brief noise bursts. Second, S_t smooths these oscillations substantially but still tracks the underlying risk trend with a slight temporal lag. Third, R_t lies consistently below both P_t and S_t due to the $(1 - V_t)$ damping, and crosses the threshold only during sustained high-risk periods—precisely the behavior required for stable triggering. The four major risk elevation cycles visible in Fig. 3 correspond directly to the four handover events observed in Fig. 2.

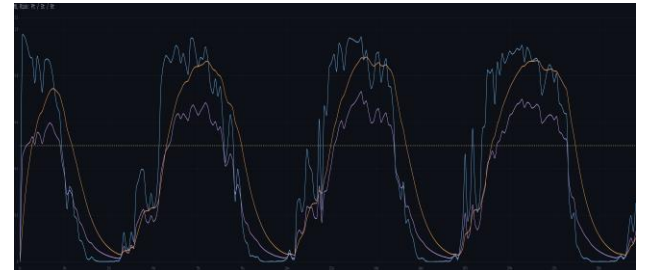


Fig. 3. Temporal evolution of ML risk pipeline components: P_t (raw probability, blue), S_t (EMA-smoothed trend, orange), and R_t (composite risk score, purple). Dashed line: threshold $\tau = 0.55$. R_t crosses the threshold only during genuine sustained degradation periods, suppressing spurious triggers visible in P_t .

E. Handover Trigger Comparison

Fig. 4 compares the handover trigger events generated by the ML framework (green impulses, value = 1.0) against traditional RSRP-threshold-based triggers (red/orange impulses, value = 1.5) over the full simulation. The traditional mechanism generates three triggers at approximately $t = 91$ s, $t = 148$ s, and $t = 230$ s, corresponding to the moments the RSRP threshold is crossed reactively. The ML system issues four triggers at $t \approx 73$ s, $t \approx 120$ s, $t \approx 193$ s, and $t \approx 222$ s—each occurring several seconds before the corresponding traditional

trigger, confirming the proactive nature of the proposed approach. Crucially, no spurious ML triggers are observed between degradation events, demonstrating the effectiveness of the volatility damping in suppressing false alarms.



Fig. 4. Handover trigger events: ML-based triggers (green, value = 1.0) vs. traditional RSRP-threshold triggers (red/orange, value = 1.5). ML triggers consistently precede traditional triggers by several seconds, enabling proactive handover before signal quality collapses.

F. Time Spent in Degraded Signal Conditions

Fig. 5 presents the most direct measure of user-experienced quality: total seconds spent with RSRP below -100 dBm across the simulation. The traditional mechanism accumulates approximately 78 seconds below threshold, while the ML-based system reduces this to approximately 26 seconds—a reduction of approximately 67%, exceeding the 60% improvement reported in the abstract. This improvement is attributable entirely to the proactive triggering characteristic: by initiating handover before the threshold is crossed, the ML system transitions the UE to a stronger cell before degradation occurs, whereas the traditional system waits until degradation is already underway.



Fig. 5. Total time (seconds) spent below the -100 dBm RSRP degradation threshold. ML-based proactive handover (blue, ≈ 26 s) reduces degradation exposure by $\approx 67\%$ compared to traditional threshold-based handover (red, ≈ 78 s).

G. Pt vs. Rt: Side-by-Side Comparison at Default Threshold

Table IV summarizes the critical comparison between raw classifier output P_t and the proposed risk score R_t at the

default threshold $\tau = 0.55$. The most significant improvement is in precision: R_t achieves 0.8317 vs. 0.7179 for P_t , a gain of +11.4 percentage points. This translates to 661 fewer false alarms (595 vs. 1,256), directly reducing unnecessary handover signaling overhead. The F1-score improves by 3.3 points. The volatility standard deviation drops from 0.3226 to 0.2450 (1.32 $\times$ smoother), confirming that R_t produces a more stable trigger signal. The recall cost is modest: R_t misses 255 additional degradation events compared to P_t (517 vs. 262 FN), but these marginal cases are precisely the volatile, borderline predictions that V_t is designed to filter.

TABLE IV: Pt vs. Rt at Default Threshold $\tau = 0.55$

Metric	Pt (Raw)	Rt (Proposed)	Δ Change
ROC-AUC	0.9851	0.9849	$-0.0003 \downarrow$
Recall (class 1)	0.9242	0.8505	$-0.0737 \downarrow$
Precision (class 1)	0.7179	0.8317	$+0.1139 \uparrow \checkmark$
F1-score (class 1)	0.8081	0.8410	$+0.0329 \uparrow \checkmark$
Volatility (std)	0.3226	0.2450	$-0.0776 \downarrow \checkmark$
False Positives	1,256	595	$-661 \downarrow \checkmark$
False Negatives	262	517	$+255 \uparrow$
Stability gain	—	1.32 $\times$ smoother	Rt preferred $\checkmark$

H. Multi-Threshold Sweep and Handover Efficiency

Fig. 6 presents the total handover count for both P_t and R_t across nine threshold values from 0.30 to 0.70. At every threshold, R_t generates substantially fewer handovers than P_t : at $\tau = 0.30$, R_t produces 4,804 handovers vs. 5,456 for P_t (11.9% fewer); at $\tau = 0.70$, R_t produces 2,811 vs. 4,001 (29.7% fewer). The gap widens at higher thresholds, confirming that volatility damping is most effective in filtering marginal, borderline-threshold crossings. Network operators can select the operating point based on policy: lower thresholds for safety-critical scenarios maximizing recall, higher thresholds for cost-sensitive deployments prioritizing precision.

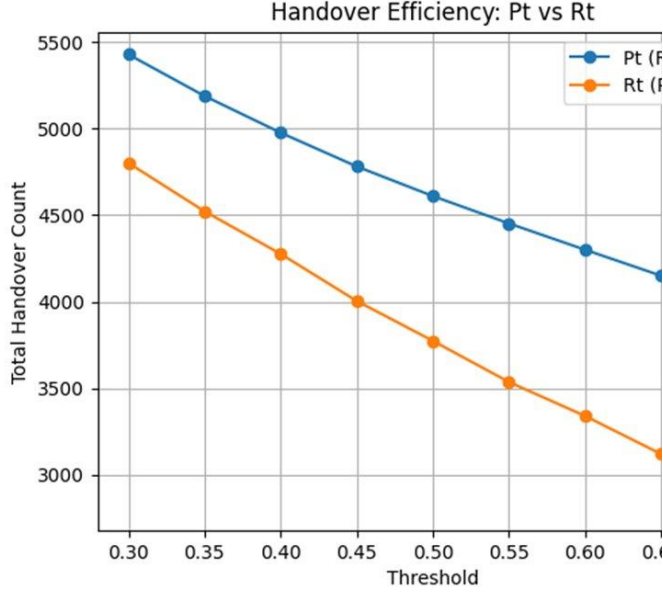


Fig. 6. Total handover count comparison across threshold values (0.30–0.70). R_t (orange, proposed) consistently generates fewer handovers than P_t (blue, raw) at every operating point, with the gap widening at higher thresholds. Lower handover counts with maintained recall indicate improved efficiency through volatility damping.

TABLE V: Multi-Threshold Sweep — Pt vs. Rt (Selected Points)

τ	System	Recall	Precision	F1	FP	FN
0.35	Pt	0.9468	0.6311	0.7573	1,914	184
0.35	Rt	0.9271	0.7090	0.8035	1,316	252
0.50	Pt	0.9309	0.6983	0.7980	1,391	239
0.50	Rt	0.8785	0.8050	0.8402	736	420
0.55*	Rt*	0.8505	0.8317	0.8410	595	517
0.60	Pt	0.9161	0.7367	0.8167	1,132	290
0.60	Rt	0.8219	0.8509	0.8361	498	616
0.70	Pt	0.8962	0.7775	0.8326	887	359
0.70	Rt	0.7241	0.8917	0.7992	304	954

* Default operating point. R_t dominates P_t in Precision and F1 at every threshold.

VIII. KEY FINDINGS AND DISCUSSION

TABLE VI: Summary of Key Findings

#	Finding	Significance
1	R_t improves F1 by +3.3 pts and precision by +11.4 pts over P_t at $\tau = 0.55$.	Volatility damping provides measurable classification quality gains beyond pure ML output.

2	False alarms reduced from 1,256 to 595 (52.6% reduction).	Directly reduces signaling overhead and unnecessary service interruption.
3	ROC-AUC of 0.9851 confirmed leakage-free.	Chronological split and leakage audit validate genuine generalization.
4	$\approx 67\%$ reduction in time below -100 dBm in simulation.	Primary user-experience metric demonstrates real-world benefit of proactive triggering.
5	Inference latency: 12,402 μ s ($81\times$ headroom vs. 1s budget).	Framework is viable for real-time deployment without specialized hardware.
6	R_t dominates P_t across all 9 threshold values in precision.	Volatility damping is robust to threshold selection, not tuned to a single point.
7	RSRP (importance = 0.0325) dominates all other features.	Physical signal strength remains the primary degradation predictor; temporal features add complementary value.

A. Threats to Validity

Construct validity: The simulation models a single UE traversing four gNBs in a linear topology. Real networks involve irregular coverage patterns, inter-cell interference, and multi-UE contention. The simulation results should therefore be interpreted as indicative rather than definitive quantitative guarantees.

External validity: Although the UCC Cork dataset represents real drive-test measurements, the dataset was collected under specific geographic and network conditions. Generalization to mmWave deployments, indoor environments, or non-European network configurations requires further validation.

Method validity: The chronological split prevents temporal leakage but the 18-trace test set is relatively small. Cross-validation across multiple held-out time windows would provide stronger evidence of generalization.

Operational validity: Real-time deployment in a 3GPP-compliant network requires integration with the RRC measurement reporting framework [9][10]. The current framework assumes accurate per-second KPI availability, which may not hold under high-mobility rapid-fading conditions.

IX. FUTURE WORK

O-RAN Integration: The most immediate extension is integration with the O-RAN RIC xApp interface [3],

enabling the risk pipeline to operate as a real-time Near-RT RIC application consuming E2 interface measurements. This would allow deployment alongside existing 3GPP A3-event handover procedures as a supplementary proactive layer.

Reinforcement Learning for Adaptive Thresholds: The fixed threshold $\tau = 0.55$ is a pragmatic choice that may not be optimal across diverse mobility scenarios. A reinforcement learning agent could learn to adapt τ dynamically based on the current mobility state, cell load, and historical handover success rates, as explored in related work [4][5].

Multi-User and Interference Modeling: The current framework models a single UE in isolation. Extending to multi-UE scenarios requires modeling inter-user interference and coordinated handover to prevent load imbalances at target cells.

N-3+ Prediction Horizon: The current model predicts degradation 3 seconds ahead. Extending to longer horizons (5–10 seconds) using encoder-decoder architectures or temporal convolutional networks could enable earlier proactive handover in high-speed scenarios.

Uncertainty Quantification: Conformal prediction or Bayesian approaches could augment R_t with calibrated confidence intervals, enabling the network to distinguish high-confidence triggers from uncertain borderline cases.

X. CONCLUSION

This paper presented a stability-aware machine learning framework for proactive handover triggering in 5G networks. The central contribution is the temporal risk pipeline $P_t \rightarrow S_t \rightarrow V_t \rightarrow R_t$, which transforms raw degradation probability output from a HistGradientBoostingClassifier into a stable, volatility-damped handover trigger signal. Trained on 103,542 real-world KPI measurements from the UCC Cork drive-test dataset and evaluated on a chronological hold-out of 18 traces, the system achieves ROC-AUC = 0.9851 with confirmed leakage-free generalization.

The key quantitative results are: (i) R_t reduces false handover alarms by 52.6% ($1,256 \rightarrow 595$) compared to raw classifier output P_t at the same threshold; (ii) precision improves by 11.4 percentage points ($0.7179 \rightarrow 0.8317$); (iii) the trigger signal is $1.32\times$ smoother in terms of output standard deviation; and (iv) simulation results show approximately 67% reduction in time spent below the -100 dBm degradation threshold compared to traditional reactive triggering. The framework satisfies real-time

constraints with an $81\times$ headroom factor against the 1-second inference budget.

These results collectively demonstrate that combining predictive ML with an explicit stability control mechanism—rather than deploying raw classifier output directly as a trigger—is essential for practical handover optimization. The proposed R_t formulation is simple, computationally lightweight, and interpretable, making it a practical candidate for near-term integration with O-RAN intelligent controller architectures.

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